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DNA Methylation Dynamics Along the Normal–Adjacent Axis in Breast Cancer

A Quantitative Statistical and Machine Learning Approach

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3.1 Methodological Objectives

This chapter establishes the methodological foundation for all subsequent analyses by addressing the construction, organization, and preliminary characterization of the DNA methylation datasets used in this thesis. Following the biological background and problem formulation introduced in Chapters 1 and 2, the present chapter acts as a bridge between domain-specific knowledge and quantitative modeling, ensuring that downstream analyses are grounded on a technically sound and well-understood data representation.

Genome-wide DNA methylation data are characterized by extremely high dimensionality, heterogeneous signal-to-noise ratios, and the coexistence of subtle biological effects with multiple sources of technical variability. In this context, premature application of statistical models or feature-selection procedures may lead to biased conclusions driven by artefacts rather than genuine biological structure. A careful examination of data quality, distributional properties, and internal consistency is therefore a necessary prerequisite for any reliable inference [1].

The primary objective of this chapter is to construct a coherent and reproducible representation of the available datasets and to assess their structural and statistical properties prior to any form of preprocessing or modeling. Specifically, this chapter aims to: (i) describe the organization of methylation matrices and associated phenotype information; (ii) characterize the global and local variability of methylation levels within each dataset; (iii) evaluate the degree of separation between Normal, Adjacent, and Tumor tissues using exploratory tools; and (iv) compare datasets at an inter-cohort level to assess consistency, heterogeneity, and potential limitations in signal transferability.

The scope of this chapter is deliberately restricted to exploratory and diagnostic analyses. No feature selection, predictive modeling, or formal statistical inference is performed at this stage. All analyses are intended to be descriptive and comparative, serving to highlight structural properties of the data and to identify potential technical or biological issues that must be addressed before proceeding further.

Methodologically, the chapter is organized around a unified exploratory framework applied consistently across all datasets. Intra-dataset analyses are used to investigate internal structure, variability patterns, and group-level behavior, while inter-dataset comparisons focus on assessing cross-cohort consistency and platform-related differences. Throughout the chapter, visualizations and summary metrics are employed as diagnostic tools rather than as decision-making instruments.

The results of this exploratory phase reveal both shared characteristics and dataset-specific behaviors, as well as clear limitations of working directly with raw or minimally processed methylation data. These observations motivate the need for a structured and rigorous data pre-processing pipeline, which is developed and justified in the following chapter. The present chapter therefore provides the conceptual and technical basis for the preprocessing choices adopted in Chapter 4 and for all subsequent modeling steps.

3.2 Dataset Construction, Metadata Integration, and Data Organization

This section provides the methodological framework for the selection, construction of the dataset, and the organisation process adopted in this study. It contextualises the data sources, representation choices, metadata management, and computational design principles that underpin the detailed implementation described in the following sections.

3.2.1 Data Sources and Study Design

All datasets analyzed in this thesis were obtained from the NCBI Gene Expression Omnibus (GEO) [2], which was selected as a unified data source to minimize heterogeneity in data formats, metadata conventions, and access mechanisms. GEO provides curated series-level accessions, stable sample identifiers, and standardized links to processed methylation matrices and sample-level annotations, making it a natural reference repository for large-scale DNA methylation studies.

The study design adopts a multi-cohort comparative framework focused on genome-wide DNA methylation in breast tissue. Three independent GEO series were selected for analysis: **GSE69914** [3], based on the Illumina HumanMethylation450 platform, and two more recent EPIC-based cohorts, **GSE225845** [4] and

GSE287331 [5]. Together, these datasets span normal/healthy, adjacent-normal, and tumor tissue states across two generations of Illumina methylation arrays (450K and EPIC), enabling both within-dataset characterization and cross-dataset comparisons.

Dataset selection was driven by strict inclusion criteria aimed at ensuring technical compatibility, biological relevance, and analytical feasibility. In particular, only studies providing processed genome-wide methylation matrices in terms of β-values were considered, allowing a uniform data representation and avoiding the need for low-level signal reprocessing from raw `.idat` intensity files. Raw `.idat` files store probe-level fluorescence intensities measured in the methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) channels, from which β-values are derived as a normalized ratio [6]:

$$\beta := \frac{\max(y^{(M)},0)}{\max(y^{(M)},0) + \max(y^{(U)},0) + \alpha}, \qquad \beta \in [0,1]. \tag{3.1}$$

where α is a small offset (typically 100) used to stabilize the denominator. Additional requirements included the use of Illumina HumanMethylation450 (450K) or MethylationEPIC (EPIC) platforms, the availability of multiple tissue classes within each cohort, and sufficiently large sample sizes to support robust statistical analysis.

3.2.2 Methylation Data Representation and Storage

In several datasets, the original authors applied established normalization procedures, including methods specifically designed to correct probe-type design bias such as BMIQ [7]. Reprocessing raw IDAT files would therefore not add information relevant to the objectives of this thesis, while substantially increasing technical complexity and reducing reproducibility. For these reasons, all analyses were conducted on the processed β-value matrices as distributed via GEO.

To support scalable analysis and efficient input/output operations, all working methylation matrices were converted to a standardized `sample` $\times$ `CpG` layout and stored in columnar Parquet format using single-precision floating-point encoding.

This choice is motivated by both numerical and computational considerations. Since β-values are strictly bounded in the interval [0,1] [6], and biologically meaningful methylation differences typically occur at magnitudes on the order of 10^{-2}–10^{-3}, single-precision floating point (`float32`, machine $\varepsilon \approx 10^{-7}$) provides more than sufficient numerical accuracy. At the same time, it substantially reduces memory usage and I/O overhead compared to double precision.

This representation is consistent with recent large-scale genomics frameworks that process molecular features, including DNA methylation data, entirely in `float32` precision for efficiency and scalability [8].

3.2.3 Phenotype Tables and Label Harmonization

For each dataset, a dedicated phenotype table was constructed by extracting and parsing GEO sample-level metadata. These tables encode tissue labels, sample identifiers, and relevant clinical or technical covariates when available. Given the heterogeneity of original annotations across studies, tissue classes were mapped to harmonized numeric encodings to ensure consistency across cohorts while preserving dataset-specific attributes in separate fields.

A strict one-to-one correspondence between methylation matrices and phenotype tables was enforced. Samples lacking either methylation measurements or corresponding metadata were excluded to prevent inconsistencies and downstream misalignment. This design guarantees that all analyses operate on synchronized molecular and phenotypic representations and enables reliable stratification by tissue class in both intra- and inter-dataset comparisons.

3.2.4 Reproducibility and Computational Workflow

The overall data organization and computational design follow a multi-dataset analytical paradigm commonly adopted in recent large-scale DNA methylation studies, in which cohort-specific analyses are complemented by cross-dataset comparisons to explicitly evaluate robustness and generalizability, while simultaneously revealing cohort-dependent effects and limitations in signal transferability that motivate subsequent methodological refinement [9].

All dataset construction and organization steps were implemented using deterministic, script-based pipelines with explicit schema definitions and typed data representations. Particular care was taken to avoid unnecessary in-memory operations on high-dimensional methylation matrices, favoring streaming ingestion, chunked processing, and out-of-core transformations when required by dataset size.

The resulting data organization yields a set of standardized, self-contained datasets, each consisting of a genome-wide methylation matrix and an aligned phenotype table stored in efficient, portable formats. This design ensures full reproducibility of the data preparation process and provides a robust foundation for the preprocessing, exploratory analysis, and modeling steps developed in subsequent chapters.

3.3 Intra-Dataset Exploration and Visualization

This section presents a structured intra-dataset exploratory analysis aimed at characterising the internal statistical and biological organisation of each methylation cohort prior to any preprocessing or modelling step. Each dataset is analysed independently in order to assess data integrity, quantify variability patterns, and

Table 3.1: Overview of the GSE69914 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	Normal (BRCA1)	Tumor (BRCA1)
485,512	407	50	42	305	3	7

evaluate whether Normal, Adjacent, and Tumor tissues exhibit distinguishable epigenetic behaviour.

In the context of breast cancer, epigenetic deregulation is known to manifest through both global and local phenomena, including genome-wide hypomethylation, focal hypermethylation of tumour-suppressor regions, and the presence of pre-neoplastic "field defects" in histologically normal tissues proximal to tumours [10, 11, 12]. These mechanisms motivate an exploratory investigation focused not only on overt tumour-associated changes, but also on more subtle alterations in Normal and Adjacent samples, which are central to the biological hypothesis of this thesis.

This section therefore applies a unified exploratory framework to each dataset (GSE69914 [3.3.1], GSE225845 [3.3.2], GSE287331 [3.3.3]), systematically examining data quality, global methylation distributions, sample-level summaries, CpG-wise variability and instability, correlation structure, dimensionality reduction, and genomic context coverage. The use of a consistent analytical protocol enables direct comparison of intra-dataset behaviour while preserving cohort-specific characteristics.

3.3.1 Dataset GSE69914

Dataset overview GSE69914 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium HumanMethylation450 BeadChip, providing genome-wide coverage of approximately 4.8×10^5 CpG loci. The dataset comprises samples spanning the full Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

An overview of the sample composition and CpG coverage is reported in Table 3.1, including the distribution across tissue classes and hereditary-risk subgroups.

A limited subset of samples is associated with germline *BRCA1* mutations. *BRCA1*, in particular, encodes a key DNA repair protein, and pathogenic variants are known to substantially increase lifetime breast and ovarian cancer risk [13]. Multiple studies have shown that BRCA-associated breast tissues exhibit distinct baseline epigenetic profiles, reflecting inherited genomic instability rather than sporadic tumorigenesis [14, 15].

Since the primary objective of this thesis is to characterise early, field defect DNA methylation alterations along the Normal–Adjacent axis in sporadic breast cancer, samples carrying *BRCA1* mutations were excluded from the core exploratory and

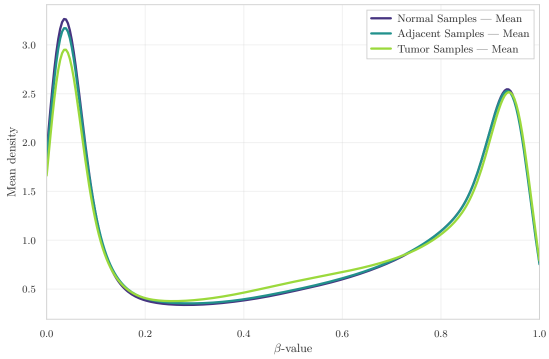


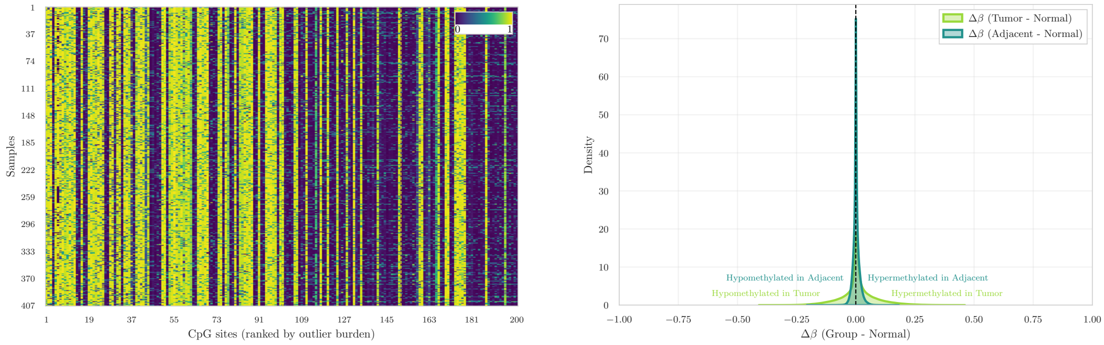
Figure 3.1: Group-wise mean β-value density in GSE69914.

preprocessing pipeline. Their inclusion would introduce a confounding hereditary epigenetic signal, potentially obscuring subtle methylation shifts in non-mutated normal and adjacent tissues.

Global methylation distributions To characterise the overall methylation landscape of the dataset, the distribution of β-values (fractional methylation in [0,1]) was first examined. The group-wise mean density curve (Figure 3.1) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [11, 16].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high-β peak and a broader low-β tail, consistent with the well-described phenomenon of global hypomethylation and increased heterogeneity in cancer [10]. Adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field defects occurring in histologically non-neoplastic tissue [12].

CpG-level instability and recurrent outliers To characterise locus-specific instability, the 200 CpG sites with the highest outlier burden across samples were examined. The heatmap of the top outlier loci (Figure 3.2a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal hypermethylation and focal hypomethylation. In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic “too much and too little methylation” behaviour of tumor genomes [10].



(a) Top outlier CpG loci.

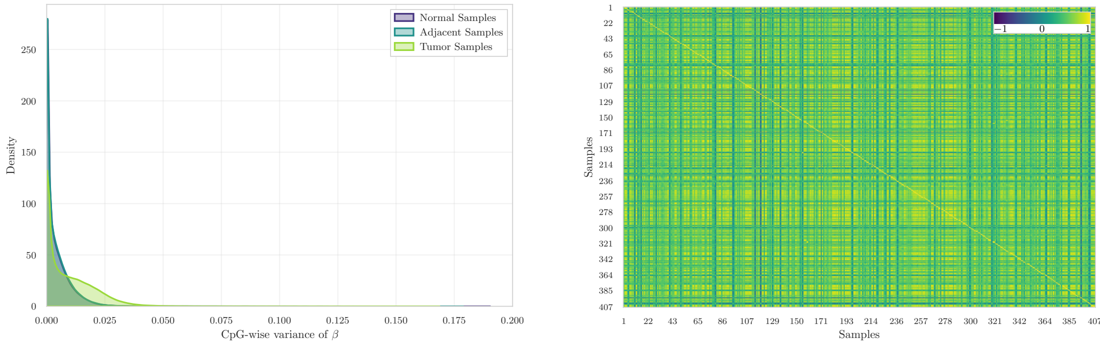
(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.2: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE69914.

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.2b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [12].

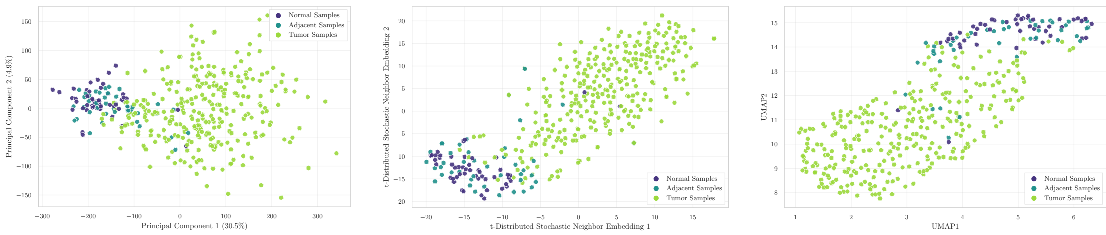
Variance and correlation structure To quantify within-group epigenetic variability, the distribution of CpG-wise variance across samples was examined. The density curves (Figure 3.3a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples.

A complementary perspective is provided by the sample correlation heatmap (Figure 3.3b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [7]. Importantly, the heatmap also reveals that all samples are highly mutually correlated, with no sharp blocks or boundaries separating the three tissue labels. This confirms that global methylation structure is largely conserved across Normal, Adjacent and Tumor tissues, and that group differences are too subtle to be detected through sample–sample correlation alone.



(a) CpG-wise variance distributions. **(b)** Sample correlation heatmap.

Figure 3.3: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE69914.

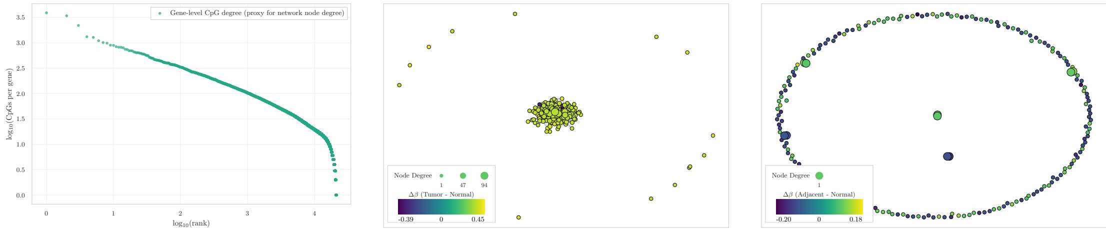


(a) PCA trustworthiness = 0.869. **(b)** t-SNE trustworthiness = 0.888. **(c)** UMAP trustworthiness = 0.874.

Figure 3.4: Low-dimensional embeddings samples in GSE69914.

Low-dimensional embeddings Dimensionality reduction was applied using Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) to visualise global similarities among samples (Figure 3.4). Across all three methods, Normal and Adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding. The trustworthiness scores are similar across embeddings (0.869–0.888), indicating that local neighbourhood relationships are largely preserved in the low-dimensional projections.

These projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are more effectively captured through locus-specific analyses rather than through global unsupervised embeddings.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.5: Dynamic-network proxy visualizations in GSE69914.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [17] manifest file, the standard Illumina resource containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the dataset exhibits the expected non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the HM450 arrays reported in the literature [18, 19]. The coverage observed is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level (Figure 3.5) provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [20, 21, 22]. Here, this conceptual framework is adapted to CpG-level methylation patterns by examining the structure of CpG–gene degree distributions and the organization of $\Delta\beta$-driven network layouts. This approach enables the exploration of whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree–rank plot (Figure 3.5a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub–periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Figure 3.5b), a compact nucleus of nodes with moderately high degree is observed, accompanied by larger $\Delta\beta$ deviations. This pattern is consistent with a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively compatible with increased network instability. In contrast, the Adjacent vs Normal network (Figure 3.5c) does not exhibit a comparable hub-like concentration and shows smaller, more diffuse deviations, suggesting that the adjacent tissue reflects only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they are conceptually consistent with the interpretation that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to a stable (pre-transition) configuration.

Overall interpretation The exploratory analysis of GSE69914 reveals a consistent but overall subtle separation between the three tissue states. Normal and Adjacent samples exhibit highly similar global methylation profiles, variance distributions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in Adjacent tissue. Across these global summaries, the two groups are so intermixed that no clear or marked distinction emerges between Normal and Adjacent samples when using whole-methylome descriptors. Tumor samples display the expected increase in heterogeneity and a shift toward hypomethylation; however, these differences remain modest at the whole-methylome level and do not produce clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs exhibit both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples together with a milder but detectable shift in Adjacent tissue. Dynamic-network proxies further indicate a more compact and perturbed core in Tumor samples, whereas Adjacent tissue appears more diffuse and weakly structured.

Overall, GSE69914 constitutes a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations are primarily localized at specific CpG loci rather than reflected in global methylome reorganization. Within this framework, the Adjacent tissue exhibits subtle yet measurable deviations from the Normal baseline, consistent with previously described field-effect patterns in breast cancer epigenomics.

3.3.2 Dataset GSE225845

Dataset overview GSE225845 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC BeadChip (850K), providing genome-wide coverage of more than 8.5×10^5 CpG loci. The dataset is part of the

Table 3.2: Overview of the GSE225845 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor
750,426	477	113	140	224

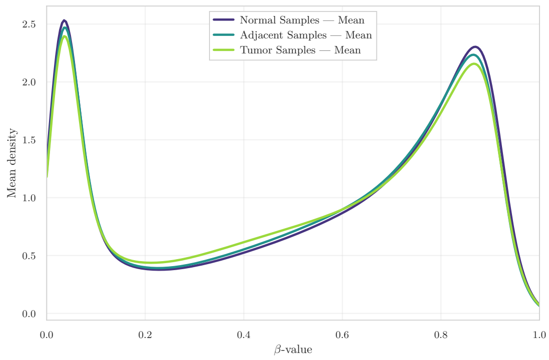


Figure 3.6: Group-wise mean β-value density in GSE225845.

NCI-Maryland Breast Cancer Cohort and includes samples spanning the Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

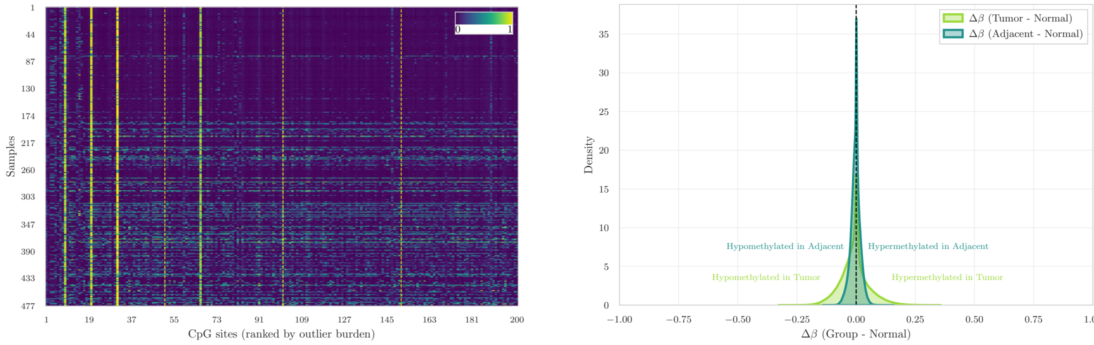
An overview of the sample composition and CpG coverage after cohort harmonization is reported in Table 3.2, including the distribution across tissue classes.

The dataset further provides detailed demographic and clinical metadata (e.g., age at surgery, race, sex, and tissue descriptors), enabling controlled downstream analyses when required.

Global methylation distributions The group-wise mean β-value densities (Figure 3.6) display the expected bimodal configuration of EPIC arrays. As observed in GSE69914, the three tissue groups show a high degree of overlap across the entire β range.

Normal and Adjacent samples exhibit nearly indistinguishable density profiles. Tumor samples present only minimal deviations, characterised by a slight increase in density at low β values and a marginal attenuation of the high-β peak. At the whole-methylome level, no marked global redistribution of methylation levels is evident.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.7a) indicates that instability within this subset is predominantly directional. Recurrent outlier events are largely associated with very low β values, consistent with focal hypomethylation affecting a restricted set of CpG sites. The perturbation pattern therefore appears concentrated rather than diffuse, suggesting localized instability rather than a genome-wide redistribution.



(a) Top outlier CpG loci.

(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.7: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE225845.

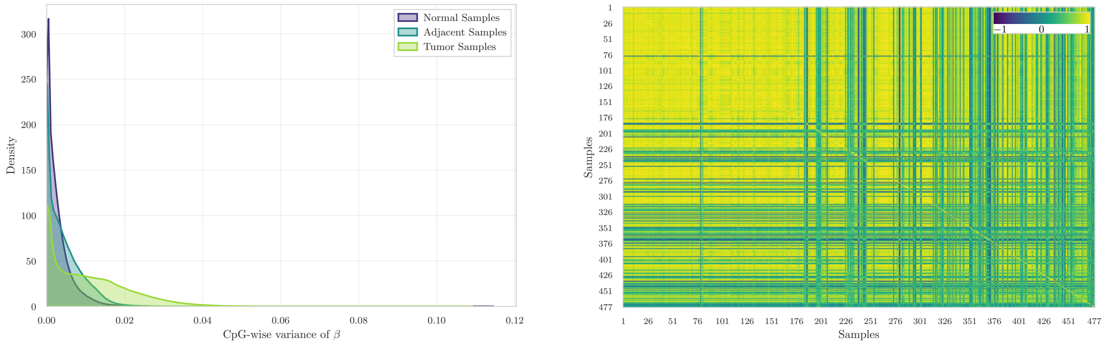
The corresponding $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.7b) are sharply centered around zero, indicating that most loci remain stable. However, both comparisons exhibit a heavier negative tail relative to the positive side. Tumor samples display the most extended negative tail, consistent with stronger hypomethylation shifts, whereas Adjacent tissue shows a milder but still detectable asymmetry toward $\Delta\beta < 0$.

Variance and correlation structure The CpG-wise variance distributions (Figure 3.8a) indicate clear differences in variability across tissue groups. Normal samples exhibit highly concentrated variance values near zero, whereas Tumor samples display a substantially longer right tail, reflecting an increased proportion of CpG loci with elevated variability. Adjacent samples show a broader distribution than Normal tissues, with partial overlap with both Normal and Tumor profiles, but without forming a strictly intermediate pattern.

The sample correlation heatmap (Figure 3.8b) shows uniformly positive correlations across the dataset. Most pairwise values lie within a narrow high-correlation range, and no distinct block structures separating tissue classes are observed. This pattern indicates preservation of global methylation structure, with group differences emerging primarily from locus-specific variability rather than from large-scale shifts in overall methylome organization.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Figure 3.9).

In PCA space, the three tissue groups largely overlap. Tumor samples show a slightly broader dispersion, particularly along the second principal component, but



(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

Figure 3.8: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE225845.

(a) PCA trustworthiness = 0.881.

(b) t-SNE trustworthiness = 0.939.

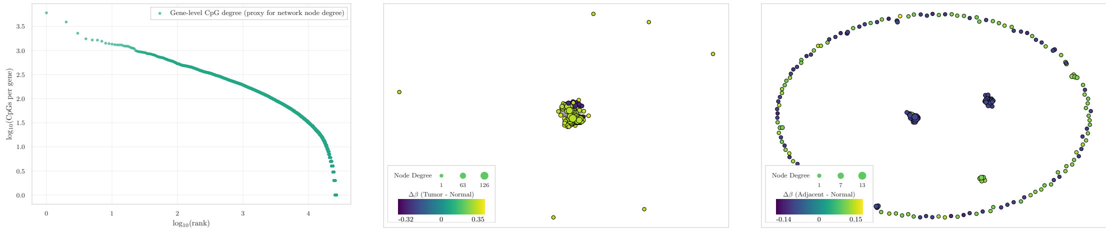
(c) UMAP trustworthiness = 0.894.

Figure 3.9: Low-dimensional embeddings samples in GSE225845.

no clear linear separation is observed.

Nonlinear embeddings reveal additional structure. In t-SNE space, Tumor samples occupy a more extended region and appear more dispersed, whereas Normal and Adjacent samples remain partially intermixed. Several Tumor samples are positioned at the periphery of the embedding, consistent with increased heterogeneity.

The UMAP projection exhibits a similar configuration. Tumor samples span a wider area and extend toward higher values of the second UMAP axis, while Normal and Adjacent samples form relatively compact yet overlapping regions. Trustworthiness scores range from 0.881 to 0.939 across embeddings, indicating good preservation of local neighbourhood structure, particularly in the nonlinear projections. Despite differences in spread, none of the methods yields sharply separated or isolated clusters.



(a) Gene-level CpG degree distribution. **(b)** DNB-like network: Tumor vs Normal. **(c)** DNB-like network: Adjacent vs Normal.

Figure 3.10: Dynamic-network proxy visualizations in GSE225845.

Genome annotation coverage Consistent with the architecture of the EPIC platform, the genomic distribution of annotated CpG contexts is markedly unbalanced. The largest fraction of probes maps to open-sea or non-island regions, which constitute the dominant category. CpG islands represent the second most abundant class, followed by north and south shores in comparable proportions, whereas north and south shelves account for the smallest fractions.

This annotation profile aligns with the expected genomic composition of the EPIC array and supports correct manifest integration.

Dynamic-network proxies The degree–rank distribution (Figure 3.10a) exhibits a heavy-tailed profile, with a limited number of genes associated with many CpGs and a long tail of low-degree nodes. Such right-skewed degree structures are characteristic of heterogeneous biological networks [23].

In the Tumor vs Normal layout (Figure 3.10b), nodes aggregate into a dense and highly compact central cluster, with only a small number of isolated peripheral points. This geometry indicates that the largest Tumor–Normal $\Delta\beta$ deviations are concentrated within a cohesive subset of CpG–gene units rather than broadly distributed across the network.

In contrast, the Adjacent vs Normal layout (Figure 3.10c) displays a markedly different configuration. While a few small central aggregates are visible, most nodes are arranged along a wide annular structure, forming a pronounced ring-like embedding. This organisation reflects weaker and more spatially dispersed deviations, without the central concentration observed in the Tumor comparison.

Altogether, the proxy visualisations indicate that Tumor–Normal contrasts generate a compact and locally coherent network signal, whereas Adjacent–Normal differences remain diffuse and geometrically fragmented.

Overall interpretation The exploratory analysis of GSE225845 indicates that, at the whole-methylome level, the three tissue states share highly similar global

Table 3.3: Overview of the GSE287331 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	OQ	CUB
866,552	446	182	60	69	67	68

methylation profiles. Group-wise mean β-value densities are nearly superimposed, and both correlation structure and linear embeddings show substantial overlap. These observations indicate preservation of large-scale methylome organisation across Normal, Adjacent, and Tumor tissues, without evidence of pronounced genome-wide redistribution of methylation levels.

Locus-focused analyses reveal more structured differences. Among the 200 CpGs with the highest outlier burden, instability is predominantly directional and characterised by focal hypomethylation. Variance distributions further show increased dispersion in Tumor samples, while Adjacent tissues exhibit broader variability than Normal without forming a strictly intermediate pattern.

Nonlinear embeddings (t-SNE and UMAP) emphasise differences in spatial dispersion rather than discrete cluster separation. Normal and Adjacent samples occupy relatively compact regions, whereas Tumor samples extend over a wider area, reflecting increased heterogeneity. Dynamic-network proxies show that Tumor–Normal contrasts generate a dense central cluster of perturbed CpG–gene units, while Adjacent–Normal differences are distributed along a broader, ring-like structure without strong central concentration.

Despite the larger sample size of this cohort, which provides stable estimates of global methylation structure, the separation between Normal and Adjacent tissues remains modest. Differences between these two groups are therefore primarily focal and do not manifest as large-scale structural shifts.

Overall, GSE225845 represents a globally homogeneous dataset in which tissue-state differences become apparent only when examined through locus-specific perturbations, variability patterns, and small-scale network organisation.

3.3.3 Dataset GSE287331

Dataset overview GSE287331 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC v1.0 BeadChip. The dataset is organised along a tumor proximity axis (TPxA) spanning multiple anatomical and pathological contexts.

Five tissue categories are represented: healthy donated breast (HDB), contralateral unaffected breast (CUB), ipsilateral opposite quadrant (OQ), adjacent normal tissue (AN), and tumor (TU). An overview of the full cohort composition is reported in Table 3.3.

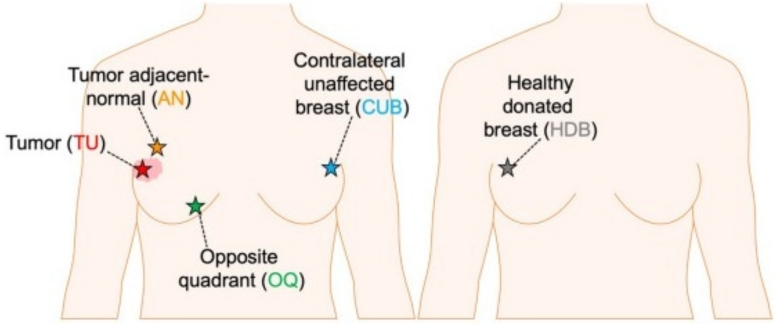


Figure 3.11: Schematic representation of the five tissue categories collected along the tumor proximity axis (TPxA), reproduced from [24].

For cross-dataset consistency, only three biologically aligned classes are retained for downstream analyses: HDB (treated as Normal baseline), AN (Adjacent), and TU (Tumor). OQ and CUB samples are excluded, as they represent intermediate benign tissues along the proximity axis and are not directly comparable with the three-class framework adopted in the other cohorts.

Global methylation distributions The group-wise mean β-value density curves (Figure 3.12) display the expected bimodal configuration of Illumina methylation arrays. A dominant peak is observed at high methylation levels, accompanied by a smaller peak consistent with standard EPIC methylation profiles [25].

Although the overall shapes remain comparable across tissue groups, a clear ordering is visible at the high-β peak. Normal samples exhibit the highest density, Adjacent samples show a slightly attenuated peak, and Tumor samples present the lowest amplitude together with a broader distribution extending toward intermediate β values. This progressive reduction in the fully methylated peak is consistent with increasing methylation variability along the Normal–Adjacent–Tumor axis.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.13a) shows that epigenetic instability is not uniformly distributed. Alterations concentrate at specific CpG sites and within specific samples, forming vertical and horizontal bands rather than a diffuse genome-wide pattern. Both directions of deviation are observed, with focal hypermethylation and focal hypomethylation present within the selected loci [10].

The corresponding $\Delta\beta$ distributions (Figure 3.13b) highlight distinct profiles for the two contrasts. The Tumor–Normal curve is broader and displays a higher density of small negative shifts, indicating widespread but moderate hypomethylation in Tumor samples. In contrast, the Adjacent–Normal distribution is more sharply centered around zero but exhibits a comparatively more pronounced negative tail,

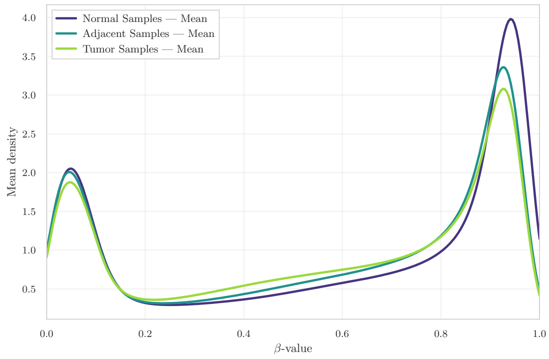
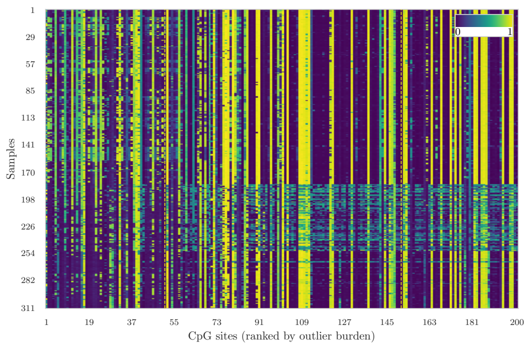
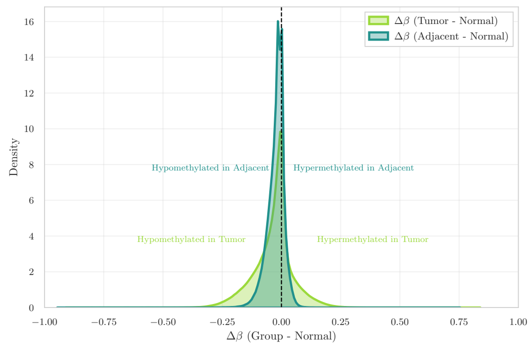


Figure 3.12: Group-wise mean β-value density in GSE287331.



(a) Top outlier CpG loci.



(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

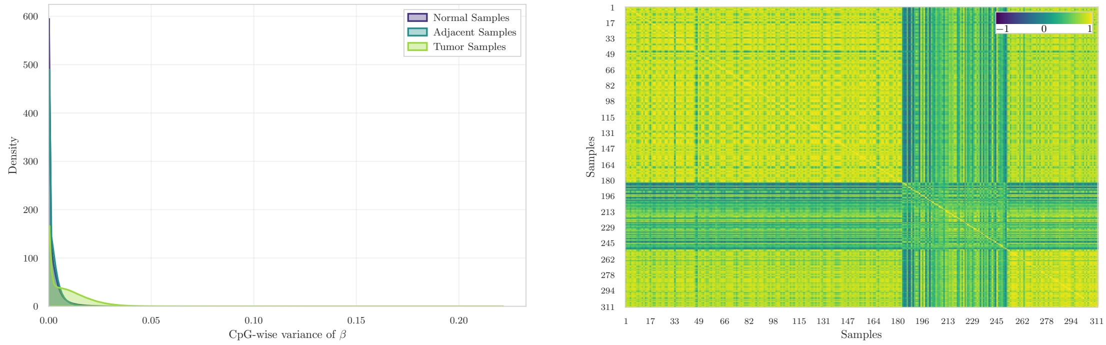
Figure 3.13: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE287331.

reflecting fewer yet larger hypomethylated deviations. Overall, hypomethylation represents the dominant direction of change in both comparisons.

Variance and correlation structure The CpG-wise variance density curves (Figure 3.14a) show that Normal and Adjacent samples are nearly superimposed, with both groups exhibiting extremely low and tightly concentrated variance values. Tumor samples display a slightly broader right tail, indicating modestly increased genome-wide variability; however, the overall distribution remains sharply peaked near zero.

At the level of CpG-wise variance, Adjacent tissue remains closely aligned with the Normal baseline, whereas Tumor samples show only a mild elevation in variability.

A complementary perspective is provided by the sample correlation heatmap



(a) CpG-wise variance distributions. **(b)** Sample correlation heatmap.

Figure 3.14: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE287331.

(Figure 3.14b). Correlations are uniformly high across the cohort, consistent with the predominance of stable CpG loci in genome-wide methylation data [7]. Although tissue labels are not overlaid on the matrix, localized blocks of higher similarity are visible, indicating structured relationships within subsets of samples rather than abrupt global separation.

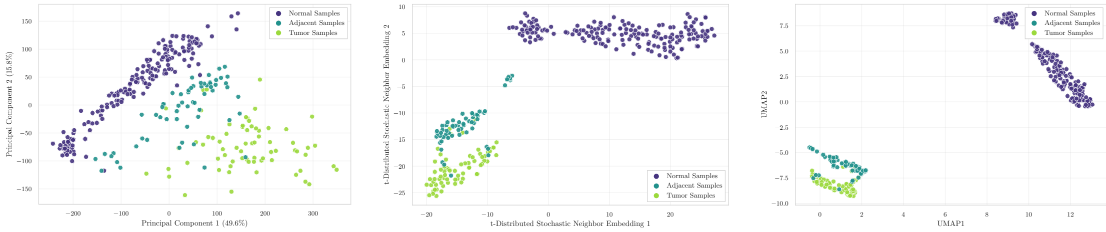
Low-dimensional embeddings Dimensionality reduction using PCA, t-SNE, and UMAP was applied to visualise the global organisation of methylation profiles (Figure 3.15).

Across all three embeddings, a consistent and well-defined structure is observed. Normal and Adjacent samples form two clearly separated clusters. The separation is visible in PCA space, becomes more pronounced in t-SNE, and appears as fully detached manifolds in the UMAP projection.

Tumor samples occupy a distinct and more dispersed region of the embedding space, reflecting increased epigenetic heterogeneity. Trustworthiness scores are uniformly high (0.961–0.980), indicating excellent preservation of local neighbourhood structure in all low-dimensional projections. No substantial overlap between Tumor and the other two groups is observed in any of the projections, indicating strong intrinsic group structure in this cohort.

Genome annotation coverage The genomic distribution of annotated CpG contexts is dominated by Open Sea regions, followed by CpG Islands, then Shores, with Shelves representing the smallest fractions.

This ordering is consistent with the established architecture of EPIC arrays [18, 19].

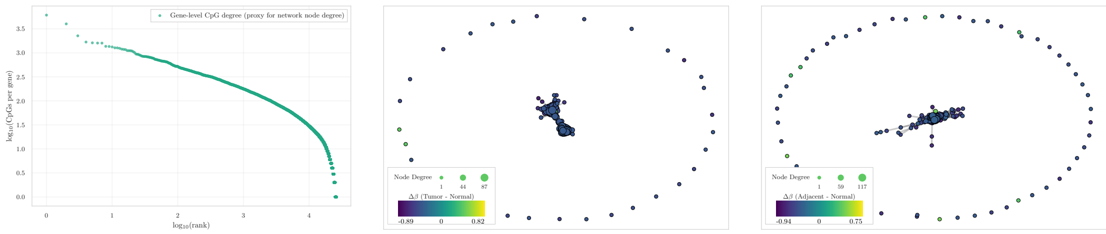


(a) PCA trustworthiness = 0.961.

(b) t-SNE trustworthiness = 0.980.

(c) UMAP trustworthiness = 0.973.

Figure 3.15: Low-dimensional embeddings of GSE287331 samples.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.16: Dynamic-network proxy visualizations for GSE287331.

Dynamic-network proxies The degree–rank curve (Figure 3.16a) shows a heavy-tailed distribution of CpG counts per gene, consistent with a hub–periphery architecture typical of biological networks.

Both $\Delta\beta$-based network layouts — Tumor vs Normal and Adjacent vs Normal (Figure 3.16b, Figure 3.16c) — share a similar overall structure: a dense and compact central block of nodes connected by short-range edges, surrounded by a ring of peripheral, weakly connected nodes. This geometry reflects the underlying degree distribution and indicates that most CpG–gene units contribute only minimally to group-level differences.

Despite this shared organisation, the Tumor vs Normal comparison exhibits a more pronounced central cluster with larger $\Delta\beta$ magnitudes, suggesting that a restricted subset of CpG–gene units undergoes locally coordinated perturbations. In contrast, the Adjacent vs Normal layout preserves the same global shape but shows weaker deviations, indicating only mild coordination of methylation changes.

Overall interpretation The exploratory analysis of GSE287331 indicates that, at the whole-methylome level, the three tissue states retain a broadly preserved global methylation architecture. Group-wise mean β-value densities follow the typical bimodal EPIC profile with only a subtle gradient (Normal > Adjacent > Tumor), and both CpG-wise variance distributions and sample correlation structure show high overall similarity across tissues, with only mild variability increases in Tumor samples.

In contrast to the previous cohorts, low-dimensional embeddings reveal a clear and consistent separation of all three groups. Normal and Adjacent samples form distinct clusters in PCA, with the separation further strengthened in t-SNE and UMAP, where the groups appear as detached manifolds. Tumor samples occupy a separate and more diffuse region, reflecting increased epigenetic heterogeneity. These projections indicate that, despite similar first- and second-order global summaries, multivariate structure in this dataset captures strong group-level organisation.

Locus-specific analyses reveal focused disruptions. Tumor samples exhibit widespread mild hypomethylation, whereas Adjacent tissue shows fewer but larger-magnitude deviations. Dynamic-network proxies further indicate a compact and perturbed central core in the Tumor comparison, while Adjacent–Normal differences display weaker coordination, consistent with early-stage methylation alterations.

Overall, GSE287331 emerges as a structurally coherent and biologically informative dataset in which Normal–Adjacent–Tumor separation is markedly more pronounced than in the other cohorts, while global methylation architecture remains broadly conserved.

3.4 Inter Dataset Exploration and Comparison

The objective of this section is not to merge the datasets—an approach that would be biologically and technically inappropriate due to differences in array platforms (HM450 vs. EPIC), preprocessing pipelines—but rather to compare their internal behaviours. Such a cross-dataset perspective enables the assessment of whether the early epigenetic alterations identified in the intra-dataset analyses are *reproducible across independent cohorts*, thereby providing support for the central biological premise of this thesis: the detection of early methylation drift in histologically normal tissues and the evaluation of its potential contribution to tumour initiation.

3.4.1 Analytical Framework

Across all three datasets, the intra-dataset analyses revealed three highly consistent phenomena:

- Normal and Adjacent tissues are globally similar, with nearly superimposed β-value distributions and comparable CpG-wise variance profiles.

Table 3.4: Cross-dataset comparison of Normal (N) vs Adjacent (A) contrasts using global and instability metrics.

Dataset	Mean $\lvert\Delta\beta\rvert$ (A–N)	Var Ratio (A/N)	Outlier Ratio (A/N)	Silhouette (A/N)
GSE69914	0.009	0.988	1.353	0.003
GSE225845	0.015	0.972	2.790	0.042
GSE287331	0.031	0.930	13.329	0.344

- Tumour samples show increased heterogeneity and a clear bias toward hypomethylation, although the magnitude of this shift varies across datasets.
- Adjacent tissues exhibit early methylation drift, detectable through $\Delta\beta$ distributions and outlier profiles even when global metrics remain highly similar to Normal.

These patterns are consistent with the expected biological progression from Normal to Tumour, in which epigenetic deregulation emerges gradually and affects normal-appearing tissue surrounding the tumour. The inter-dataset analysis therefore evaluates the *directional consistency* of these effects rather than aiming at strict quantitative comparability.

3.4.2 Cross-Dataset Comparative Metric

To quantify the epigenetic deviation of Adjacent tissue from the Normal methylome across the three cohorts, a set of global and instability-oriented metrics was computed independently within each dataset for the Adjacent versus Normal contrast.

All metrics reported in Table 3.4 were computed using the complete CpG set available in each dataset, rather than the cross-platform intersection. The resulting values therefore reflect cohort-specific dimensionality and internal variance structure.

Collectively, these metrics indicate the presence of early epigenetic drift in histologically non-tumour tissue. Across cohorts, increasing mean $|\Delta\beta|$, progressive variance imbalance, elevated outlier ratios, and improved silhouette separation consistently support a directional shift from Normal to Adjacent states, in agreement with the field-defect framework described by Teschendorff et [12] and reinforced in other tissue contexts.

Mean absolute methylation difference For each CpG, the absolute mean difference between Adjacent and Normal was defined as

$$|\Delta\beta_i| = \left|\beta_i^{(A)} - \beta_i^{(N)}\right|. \tag{3.2}$$

The cross-cohort pattern reveals progressively larger locus-specific shifts from GSE69914 to GSE225845 and GSE287331. Although the absolute magnitude of the effect remains modest, it is consistent in direction across cohorts. This monotonic increase reflects a strengthening early deviation in Adjacent tissues. Comparable locus-wise methylation drifts in histologically normal tissue have been reported in independent systems, such as colonic mucosa adjacent to adenomas [26], supporting the interpretation of these alterations as early manifestations of field cancerization.

Global variance ratio For each sample s, the global methylation variance was computed across all CpG sites: $v_s = \mathrm{Var}(\beta_{s,1}, \beta_{s,2}, \ldots, \beta_{s,m})$, where m denotes the number of CpGs. Group-wise mean variances were then defined as

$$\overline{v}^{(N)} = \frac{1}{|N|} \sum_{s \in N} v_s, \qquad \overline{v}^{(A)} = \frac{1}{|A|} \sum_{s \in A} v_s, \tag{3.3}$$

and the reported metric corresponds to their ratio:

$$\frac{\mathrm{Var(A)}}{\mathrm{Var(N)}} = \frac{\overline{v}^{(A)}}{\overline{v}^{(N)}}. \tag{3.4}$$

Across cohorts, this ratio remained consistently close to unity, indicating that Adjacent tissue does not exhibit diffuse genome-wide instability. Such behaviour suggests that early pre-neoplastic alterations are predominantly *focal* rather than global, with only specific CpG sites displaying abnormal dispersion while overall variance remains largely preserved [26].

This interpretation is consistent with contemporary models of early tumourigenesis, in which subtle and locally restricted perturbations accumulate prior to large-scale chromatin deregulation and overt neoplastic transformation [27].

Outlier burden ratio For each sample, the outlier burden is defined as the number of CpG sites whose methylation deviates from the Normal reference distribution by more than three times the Normal interquartile range:

$$\mathrm{outlier}_s = \#\left\{ i \ : \ \left|\beta_{i,s} - \tilde{\beta}_i^{(N)}\right| > 3 \cdot IQR_i^{(N)} \right\}, \tag{3.5}$$

where $\tilde{\beta}_i^{(N)}$ and $IQR_i^{(N)}$ denote the median and interquartile range of CpG i across Normal samples.

The reported A/N ratio corresponds to the mean outlier burden in Adjacent tissue divided by the mean burden in Normal tissue. Across cohorts, this metric exhibits the most pronounced cross-dataset gradient, with progressively stronger enrichment of outlier events in Adjacent tissue. Such behaviour mirrors evidence that increased local methylation variability in histologically normal tissue represents a sensitive marker of early instability [26], and provides quantitative support for the presence of marked field effects in GSE287331.

Silhouette score For each sample s, let $a(s)$ denote the mean distance from s to all other samples within the same group (cohesion), and let $b(s)$ denote the smallest mean distance from s to samples in the alternative group (separation). The silhouette coefficient for sample s is defined as

$$\mathrm{sil}(s) = \frac{b(s) - a(s)}{\max\{a(s),\, b(s)\}}, \tag{3.6}$$

and the reported score corresponds to the mean silhouette computed across all Normal and Adjacent samples.

Higher values indicate clearer multivariate separation between Normal and Adjacent states, whereas values near zero reflect substantial overlap. Across cohorts, the silhouette analysis reveals limited separation in GSE69914 and GSE225845, consistent with minimal global displacement in multivariate space. In contrast, GSE287331 exhibits partial separation, indicating that the epigenetic landscape of Adjacent tissue is measurably shifted relative to Normal. This behaviour aligns with mechanistic models in which epigenetic dysregulation constitutes one of the earliest events in tumour initiation [27], preceding morphological transformation.

Collectively, the set of metrics delineates a consistent Normal–Adjacent gradient across datasets. Early epigenetic drift is detectable in all cohorts, while its magnitude, focality, and multivariate visibility progressively intensify from GSE69914 to GSE287331. Importantly, the combination of stable global variance with pronounced outlier enrichment supports a model in which early pre-neoplastic changes arise through *localised* perturbations, a hallmark of field cancerization [26, 12].

3.4.3 Visual Comparative Analysis

To complement the quantitative summary reported in Table 3.4, this section provides a visual characterisation of the Normal–Adjacent contrast across the three cohorts. All analyses are performed on the three-way intersection of 326,330 CpG sites, ensuring maximal comparability across platforms (HM450 and EPIC) and preprocessing pipelines [18]. The objective is not to achieve numerical concordance at the single-CpG level—a notoriously difficult task due to platform differences, probe-design effects, and cohort-specific processing [19, 28]—but rather to evaluate whether the *shape*, *direction*, and *global behaviour* of early methylation drift remain coherent across studies.

Global $\Delta\beta$ distributions The overlaid $\Delta\beta$ (Adjacent–Normal) density curves across the three cohorts are shown in Figure 3.17. All distributions are sharply centred around zero, indicating that the N–A methylation drift remains globally subtle in each dataset, consistent with the focal and low-amplitude nature of early

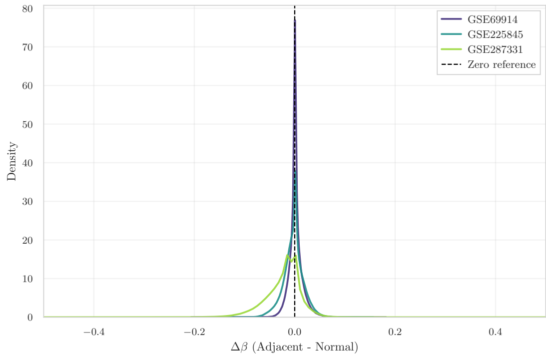


Figure 3.17: Overlay of $\Delta\beta$ (Adjacent–Normal) density curves across the three datasets, computed on the 326,330 CpG sites shared by all cohorts.

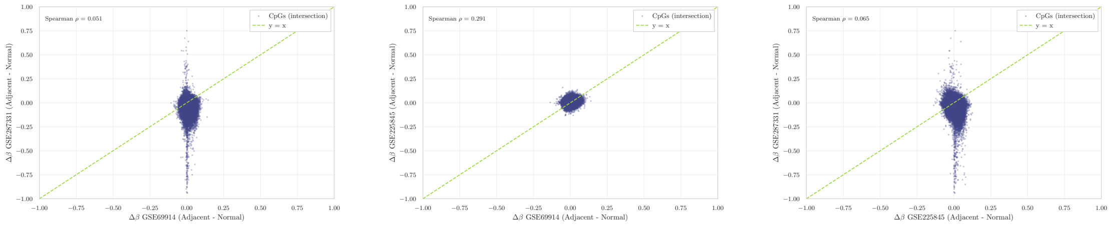
epigenetic alterations described in pre-neoplastic tissues [12, 29]. Despite differences in dispersion—reflecting platform-dependent noise, probe-type composition, and study-specific variability [16, 19]—the three curves exhibit a highly similar shape. This pattern indicates that early epigenetic deviations in Adjacent tissue, although small in magnitude, exhibit reproducible global distributional features across independent cohorts, rather than exact concordance at the single-CpG level.

Pairwise $\Delta\beta$ concordance across datasets To assess whether locus-specific A–N shifts replicate across cohorts, pairwise scatterplots are shown for all dataset pairs (Figure 3.18a, Figure 3.18b, Figure 3.18c). Each point corresponds to a CpG in the three-way intersection; axes represent the mean $\Delta\beta$ (A–N) computed independently within each dataset. Across all pairs, Spearman correlations remain weak ($\rho = 0.05$–0.29), consistent with the well-established difficulty of achieving single-CpG reproducibility across independent methylation studies [18, 28, 30].

The comparison between GSE69914 and GSE225845 exhibits mild monotonic agreement, whereas pairs involving GSE287331 show near-zero correlation. These results indicate that while the *global* N–A drift remains consistent across cohorts (as demonstrated by the density curves), *fine-scale, CpG-specific* effects are strongly influenced by platform architecture, batch structure, and study-specific variability [19, 28].

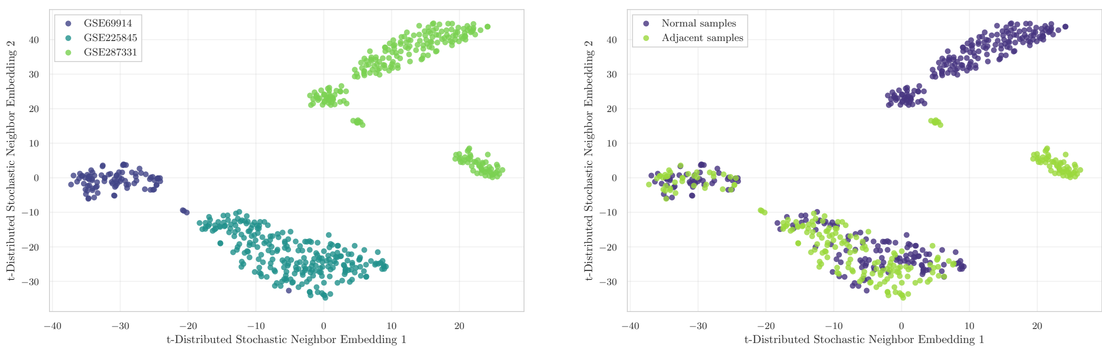
t-SNE embeddings on the shared CpG intersection To investigate the multivariate structure of the N–A contrast across cohorts, t-SNE embeddings are computed on M-values restricted to the shared CpGs. Two complementary visualisations of the embedding are provided in Figure 3.19a (coloured by dataset) and Figure 3.19b (coloured by tissue state).

When coloured by dataset, the embedding exhibits clear clustering by study,



(a) GSE69914-GSE287331, Spearman $\rho \approx 0.051$ **(b)** GSE69914-GSE225845, Spearman $\rho \approx 0.291$ **(c)** GSE225845-GSE287331, Spearman $\rho \approx 0.007$

Figure 3.18: Pairwise $\Delta\beta$ (Adjacent–Normal) scatterplots across the three methylation datasets, computed on the 326,330 shared CpG sites. Spearman ρ values are reported in each panel.



(a) t-SNE coloured by dataset trustworthiness = 0.976 **(b)** t-SNE coloured by tissue state trustworthiness = 0.976

Figure 3.19: t-SNE embeddings computed on the 326,330 CpGs shared across GSE69914, GSE225845, and GSE287331.

reflecting the dominant influence of batch and platform effects in high-dimensional methylation data [28, 30]. This pattern indicates that the global methylation structure is largely driven by inter-cohort variability, supporting the decision not to merge datasets at the raw feature level.

When coloured by tissue state (Normal vs. Adjacent), no global separation across cohorts emerges, as the embedding remains primarily structured by dataset identity. However, within individual study-specific clusters, a state-dependent displacement becomes visible. This separation is particularly evident in GSE287331, where Normal and Adjacent samples form more clearly distinct substructures. This behaviour is consistent with early, localised methylation drift, mirroring the quantitative metrics reported in Table 3.4.

Overall, the visual analyses corroborate the quantitative metrics: early epigenetic drift is detectable in all cohorts, but its magnitude and visibility vary substantially, with GSE287331 exhibiting the clearest field-defect signature. The combination of globally subtle yet directionally consistent $\Delta\beta$ patterns, weak CpG-level concordance, and dataset-specific multivariate structure reinforces the view that the Normal$\rightarrow$Adjacent transition reflects genuine early epigenetic deregulation rather than dataset-specific artefacts.

3.5 Synthesis of Exploratory Findings and Pre-Processing Rationale

Taken together, the cross-dataset analyses indicate that the epigenetic deviations observed in Adjacent tissue are not idiosyncratic artefacts of a single cohort, but represent coherent and reproducible signals whose *direction* is consistent across studies. At the same time, the results clarify that this Normal–Adjacent drift is globally subtle at the whole-methylome level, becoming visible primarily through locus-specific perturbations, focal outlier enrichment, and dataset-dependent multivariate displacements.

A key methodological outcome is the separation between global distributional coherence and weak CpG-level concordance across cohorts. When restricted to the shared CpG intersection, the Adjacent–Normal $\Delta\beta$ distributions retain a similar shape and remain sharply centred around zero, whereas pairwise concordance of locus-specific shifts is low. Together with the dominant clustering by dataset identity in shared-CpG embeddings, this confirms that array generation, cohort-specific processing, and batch structure induce strong technical signatures; consequently, merging cohorts at the raw feature level would not be biologically meaningful and would risk confounding downstream analyses.

The comparison further shows that the *strength* of the Normal–Adjacent signal differs substantially across datasets. In GSE69914 and GSE225845 the drift is detectable but modest and mainly expressed as focal instability patterns, whereas GSE287331 exhibits a markedly clearer field-defect signature, including stronger outlier enrichment and more visible separation in multivariate space. This heterogeneity motivates a preprocessing strategy that is both rigorous and conservative, aiming to reduce technical variability and probe-level artefacts without suppressing the subtle biological differences that define the Normal–Adjacent axis.

These observations provide the rationale for the structured data pre-processing pipeline developed in Chapter 4. The next chapter implements reproducible and platform-aware preprocessing steps to stabilise the methylation representation, control technical effects, and prepare the data for downstream statistical analysis and learning-based models focused on early epigenetic alterations.

Chapter 4

Data preprocessing

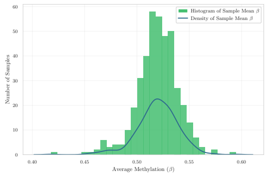
REPORT 02

Missingness filtering and imputation I quantified the amount and structure of missing values in the β-matrix. Probe-wise missingness is a well-known source of unreliability in Illumina methylation arrays, and several studies recommend removing CpGs that exhibit low measurement quality or an excessive proportion of missing values. For example, Islam *et al.* [31] excluded probes with missing β-values in more than 2% of samples in their pediatric tissue methylation analysis, while Mansell *et al.* [32] applied a similar 1% threshold for detection failure in their EPIC-array quality control pipeline.

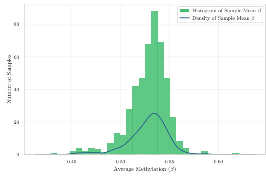
Guided by this evidence and the large number of CpG sites available, I adopted a stringent probe-wise filtering criterion and removed all CpGs missing in more than 1% of samples. This choice prioritises high-confidence measurements while minimising reliance on imputed values.

After filtering, only a negligible fraction of values remained missing in the working matrix (58 561 NaN, corresponding to 0.0187% of all entries). Given this extremely low level of sparsity, median-based imputation is both statistically robust and distribution-preserving. Following current recommendations, where missing methylation values are replaced by probe-wise medians to avoid introducing group-specific bias, all residual NaN were imputed using the median β-value of each CpG across samples. This approach is consistent with recent harmonization methods such as mLiftOver, where missing methylation values are explicitly replaced using probe-wise median substitution to preserve biologically plausible levels [33].

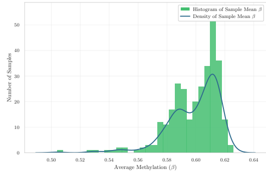
Evidence of Missing Infinium Type I/II Probe Bias Correction During the exploratory analysis performed uniformly across all three datasets, the per-sample mean β-value distributions (Fig. 4.1a, Fig. 4.1b, Fig. 4.1c) revealed a marked



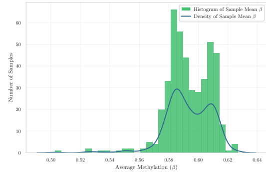
(a) GSE69914



(b) GSE225845



(c) GSE287331 pre BMIQ



(d) GSE287331 post BMIQ

Figure 4.1: Per-sample mean β distributions for the three datasets. `GSE69914` and `GSE225845` show a single broad peak, whereas `GSE287331` displays a right-shifted bimodal pattern. After BMIQ, the two peaks become slightly closer and more similar in height, but the bimodality persists.

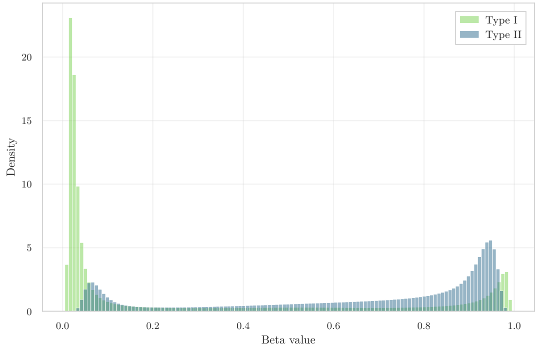
inconsistency. Whereas `GSE69914` and `GSE225845` displayed a single broad peak of global methylation levels, `GSE287331` showed a much narrower distribution, shifted towards higher β-values and characterized by two distinct modes in the high-β range.

This unusual right-shifted bimodality—largely absent in the other datasets—strongly indicated that a key preprocessing step, namely Infinium Type I/II probe-design bias correction, had not been applied in the processed data released by the authors.

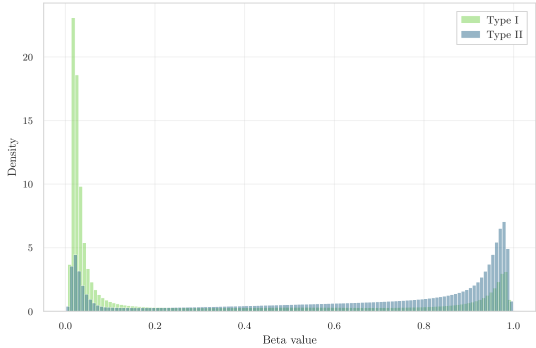
To confirm this hypothesis, I stratified all probes by Infinium design type (Type I vs. Type II) using the official Illumina EPIC v1.0 manifest, obtained from the corresponding Bioconductor annotation package [34]. Using this manifest, the β-value distributions were separated by probe design and inspected through a dedicated diagnostic plot (Fig. 4.2). According to the expected behaviour described by Teschendorff et al. [7], datasets that have undergone BMIQ normalization should display nearly overlapping β-value distributions for the two probe types.

In contrast, the Q–Q plot comparing Type I and Type II probes revealed a markedly elevated median absolute deviation (median $|\Delta| = 0.380$), indicating a strong and systematic quantile-wise shift between the two designs. Such a deviation is fully incompatible with BMIQ-corrected data, which should instead exhibit deviations close to zero across all quantiles. After applying BMIQ, this deviation decreases only marginally ($|\Delta| : 0.380 \to 0.370$), confirming that the correction improves probe-type alignment but only to a limited extent.

This modest improvement is also clearly visible in the diagnostic density plots (Fig. 4.2b). After BMIQ, the rightmost portion of the distribution (near $\beta \approx 1$) shows Type II probes becoming visibly more similar to Type I, and the strong asymmetry between the two curves is partially reduced. Nevertheless, the overall shapes of the two probe distributions remain distinct, and the characteristic "double-tail" pattern of the uncorrected dataset is still present after normalization. BMIQ



(a) GSE287331 pre BMIQ



(b) GSE287331 post BMIQ

Figure 4.2: Type I vs. Type II probe design comparison before and after BMIQ. After correction, Type II probes become more similar to Type I in the high-β region (especially near $\beta \approx 1$), and the overall discrepancy is reduced ($|\Delta| : 0.380 \to 0.370$), though probe types remain clearly distinct.

therefore warps the distribution in the expected direction, but the magnitude of the adjustment is small and insufficient to reconcile the two probe types to the degree observed in datasets processed with a full bias-correction pipeline.

A similar behaviour can be observed in the per-sample mean β distributions: after BMIQ, the two peaks become slightly higher and closer to one another (Fig. 4.1d), indicating that the algorithm partially reduces the gap between the two dominant modes. However, even after correction, the distribution remains visibly and consistently *bimodal*, suggesting that BMIQ mitigates only weakly the underlying discrepancy and that sample-level heterogeneity continues to dominate the global methylation structure.

Taken together, these observations provide clear visual evidence that no probe-design bias correction was applied in the original processing. To further support this conclusion, I examined the official publication [24]. The paper carefully describes the acquisition, processing, and analytical use of Illumina EPIC β-values, but it does *not* mention BMIQ, SWAN, noob, or any other method aimed at correcting Infinium probe-type bias. The absence of such a statement—combined with the strong visual mismatch between Type I and Type II probe distributions—confirms that no probe-design bias correction was applied before the release of the dataset.

For these reasons, applying a dedicated Infinium Type I/II bias correction step is necessary before performing any comparative analysis.

BMIQ correction: implementation and methodological considerations
Given the clear evidence of missing Infinium Type I/II probe-bias correction, I applied the canonical BMIQ algorithm [7], using the implementation provided

by the `wateRmelon` package [35], to harmonize the distributions of Type I and Type II probes for each sample. To ensure full reproducibility and strict memory control on the extremely wide EPIC matrix, I implemented a *sequential, sample-wise BMIQ correction*, avoiding any between-sample pooling or multi-core parallelization. Each sample's β vector was processed independently using a safe wrapper around `wateRmelon::BMIQ`, which first verified the availability of a sufficient number of non-missing Type I and Type II probes and then returned the corrected β-values. The corrected matrix was updated in place, preserving the schema `id_tissue | CpGs... | label` and subsequently written to a single Parquet file. This design guarantees that no cross-sample information influenced the correction and that the warping applied by BMIQ reflects exclusively the distributional characteristics of each individual sample.

While BMIQ is widely used to mitigate probe-design bias, multiple studies have raised important methodological caveats regarding normalization procedures that operate on the full distribution of methylation values. Yousefi *et al.* [36] highlighted that quantile-based adjustments and related normalization strategies may "come at the cost of *artificially reduced biological variability*," a phenomenon that is particularly undesirable when studying biological groups expected to differ globally in their methylation profiles. Similarly, Liu *et al.* [37] reported that, when methylation alterations are extensive in size and number, certain normalization methods can "*remove biological signal*," effectively shrinking meaningful differences between groups.

More recently, Welsh *et al.* [38] performed a systematic evaluation of normalization methods for EPIC arrays and showed that approaches enforcing strong distributional alignment may distort real biological structure. In particular, the authors observed that quantile-based normalization "*forces the empirical marginal distributions of the samples to be the same, removing all variation in this statistic*," and reported that BMIQ, in some settings, yielded "*distorted and discontinuous*" distributions. Such findings provide a conceptual explanation for the behaviour observed: although BMIQ was applied strictly per-sample, its within-array warping can attenuate global differences between Normal, Adjacent, and Tumor samples when these differences are manifested as broad shifts in the underlying β-value distribution.

Together, these considerations emphasise both the necessity and the limitations of probe-design bias correction. Applying BMIQ was essential to remove the technical Type I/II bias absent in the original processing, yet its known tendency to reduce distributional heterogeneity across groups must be carefully taken into account when interpreting group-level contrasts in downstream analyses.

Importantly, the modest degree of probe-type harmonization is accompanied by a noticeable flattening of biological structure: as will be shown in the low-dimensional embedding analysis (Section 3.3.3), PCA, UMAP, and t-SNE projections reveal that the Normal, Adjacent, and Tumor groups become *less clearly separable* after BMIQ. For this reason, all downstream analyses are carried out on the **non-BMIQ version** of the dataset, which preserves sharper cluster boundaries and better reflects the underlying biological signal.

This strong separation is also quantitatively confirmed by silhouette analysis: the average silhouette coefficient *before* BMIQ was $\mathrm{sil}_{\mathrm{pre}} = 0.2876$, but *after* BMIQ correction it declined to $\mathrm{sil}_{\mathrm{post}} = 0.2537$. Thus, BMIQ *reduced*–even if not significantly–the natural clusterability of the data, collapsing genuine biological differences—particularly the clear separation between Normal and Adjacent samples.

For this reason, and in accordance with the principle of preserving authentic biological signal, the BMIQ-normalised version was not adopted.

Chapter 5

Feature selection

REPORT 03

Chapter 6

Model training and evaluation

REPORT 04

Chapter 7

Dynamic System for biomarker detection

REPORT 05

Chapter 8

Conclusion and future work

CONCLUSIONI E PROSPETTIVE DI FUTURO

Chapter 9

Appendix

AGGIUNGERE APPENDICE

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